

STELLA Lab Spec Main Board

Step	Action	What value? / Note	Where? / Note	What color?	Done
1	Get board	Adafruit Permaproto Half Size		White	☐
2	Cut traces:		Rear		☐
2.01	Cut trace		30A / 30B		☐
2.02	Cut trace		30B / 30C		☐
2.03	Cut trace		30C / 30D		☐
2.04	Cut trace		30H / 30i		☐
2.05	Cut trace		30i / 30J		☐
2.06	Cut trace		29A / 29B		☐
2.07	Cut trace		29B / 29C		☐
2.08	Cut trace		29C / 29D		☐
2.09	Cut trace		29H / 29i		☐
2.10	Cut trace		29i / 29J		☐
3	Install magnetic couplers:		Front, use fixtures TBD		☐
3.1	Install magnetic coupler	4 pos RA, w spring pins	30: A, B, C, D		☐
3.2	Install magnetic coupler	3 pos RA, w spring pins	30: H, i, J		☐
4	Install bus wires:		Front		☐
4.1	3V		A29-30 to 3V (+ bus below A)	+3.3V	☐
4.2	GND		B29-30 to GND (- bus below A)	GND	☐
4.3	SCL		C29-30 to SCL	i2c Serial Clock	☐
4.4	SDA		E30 to SDA	i2c Serial Data	☐
4.5	5V		J29-30 to G2	5V	☐
5	Install transistors:	PN2222ABU	Front: Flat face towards J: Solder the center pin first, then straighten the body before soldering the other two pins.		☐
5.1	Install transistor	PN2222ABU	B12-13-14		☐
5.2	Install transistor	PN2222ABU	G12-13-14		☐
5.3	Install transistor	PN2222ABU	B16-17-18		☐
5.4	Install transistor	PN2222ABU	G16-17-18		☐
6	Install shunt resistors:	R1Ω 1% tolerance 1/4W	Front		☐
6.1	Install shunt resistor	R1Ω	B21 to GND		☐
6.2	Install shunt resistor	R1Ω	B22 to GND		☐
6.3	Install shunt resistor	R1Ω	B23 to GND		☐
6.4	Install shunt resistor	R1Ω	B24 to GND		☐
7	Install pull down resistors:	100kΩ	Front		☐
7.1	Install pull down resistor	100kΩ	B5 to GND		☐
7.2	Install pull down resistor	100kΩ	B6 to GND		☐
7.3	Install pull down resistor	100kΩ	B7 to GND		☐
7.4	Install pull down resistor	100kΩ	B8 to GND		☐
8	Install bias limit resistors:	R50Ω	Front		☐
8.1	Install bias limit resistor	R50Ω	C11-E13		☐
8.2	Install bias limit resistor	R50Ω	H11-J13		☐
8.3	Install bias limit resistor	R50Ω	C15-E17		☐
8.4	Install bias limit resistor	R50Ω	H15-J17		☐
9	Install socket headers: Front		Front		☐
9.1	Install socket header	6 pos	D4-D9		☐
9.2	Install socket header	6 pos	G4-G9		☐
9.3	Install socket header	6 pos	D20-D25		☐
9.4	Install socket header	6 pos	G20-G25		☐
10	Install socket headers: Rear		Rear		☐
10.1	Install socket header	3x3 (9 pos)	A,B,C x 1,2,3		☐
10.2	Install socket header	3x3 (9 pos)	H,i,J x 1,2,3		☐
10.3	Install socket header	2x3 (6 pos)	GND 1,2,3 and 3V 1,2,3		☐
10.4	Install socket header	1x3 (3 pos)	SDA 1,2,3		☐
11	Install ground wires:		Front	GND	☐
11.1	Install ground wire		B9 to GND	GND	☐

Step	Action	What value? / Note	Where? / Note	What color?	Done
11.2	Install ground wire		F24 to GND	GND	☐
12	Install 3V wires:		Front	+3.3V	☐
12.1	Install 3V wire		B4 to 3V	+3.3V	☐
12.2	Install 3V wire		F25 to 3V	+3.3V	☐
13	Install SCL wires:		Front	i2c Serial Clock	☐
13.1	Install SCL wire		G3 to SCL	i2c Serial Clock	☐
13.2	Install SCL wire		J7 to SCL	i2c Serial Clock	☐
13.3	Install SCL wire		J23 to SCL	i2c Serial Clock	☐
14	Install SDA wires:		Front	i2c Serial Data	☐
14.1	Install SDA wire		J6 to SDA	i2c Serial Data	☐
14.2	Install SDA wire		J22 to SDA	i2c Serial Data	☐
15	Install channel 0 wires:		Front	CH0	☐
15.1	Install channel 0 wire	LED lamp low to collector	F1 to E12	CH0	☐
15.2	Install channel 0 wire	DAC to base through R_limit	C8 to A11	CH0	☐
15.3	Install channel 0 wire	emitter to R_shunt at ADC	A14 to C21	CH0	☐
16	Install channel 1 wires:		Front	CH1	☐
16.1	Install channel 1 wire	LED lamp low to collector	E1 to A16	CH1	☐
16.2	Install channel 1 wire	DAC to base through R_limit	C7 to A15	CH1	☐
16.3	Install channel 1 wire	emitter to R_shunt at ADC	E18 to E22	CH1	☐
17	Install channel 2 wires:		Front	CH2	☐
17.1	Install channel 2 wire	LED lamp low to collector	E2 to F12	CH2	☐
17.2	Install channel 2 wire	DAC to base through R_limit	E6 to F11	CH2	☐
17.3	Install channel 2 wire	emitter to R_shunt at ADC	F14 to E23	CH2	☐
18	Install channel 3 wires:		Front	CH3	☐
18.1	Install channel 3 wire	LED lamp low to collector	E3 to F16	CH3	☐
18.2	Install channel 3 wire	DAC to base through R_limit	E5 to F15	CH3	☐
18.3	Install channel 3 wire	emitter to R_shunt at ADC	F18 to E24	CH3	☐
19	Test board for continuity and shorts				☐
20	Install MCP4728 12 bit DAC		D4-D9; Vcc, VD, VC, VB, VA, GND G4-G9: RDY, LDAC, SDA, SCL, GND, Vcc		☐
21	Install ADS1015 12 bit ADC		D20-D25; A-, A0, A1, A2, A3, A+ G20-G25; ALRT, ADDR, SDA, SCL, GND, Vin		☐
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	reference drawing file:	lab spec main board.PDF			